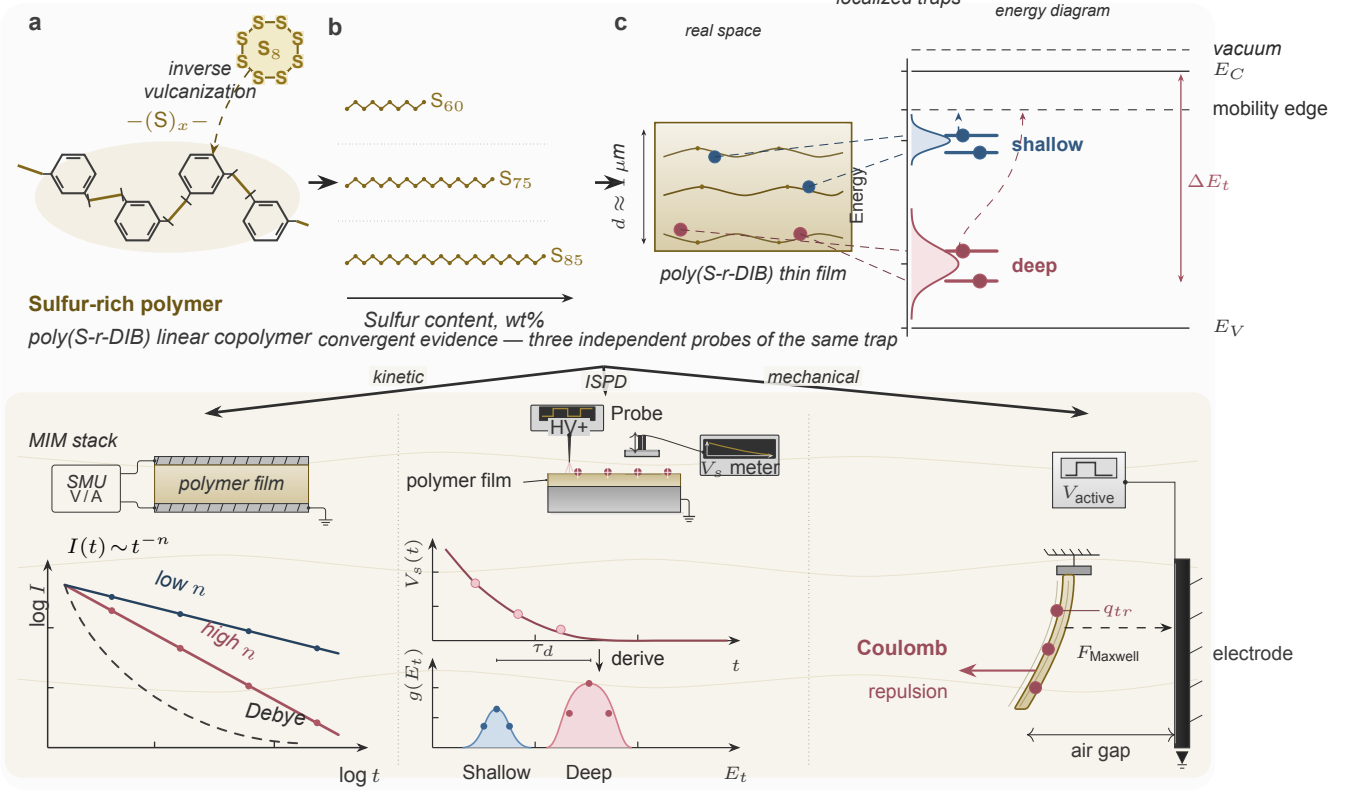
a
b
c
inverse vulcanization
$-(S)_x-$
S_8
S_{60}
S_{75}
S_{85}
Sulfur-rich polymer
poly(S-r-DIB) linear copolymer
Sulfur content, wt%
real space
localized traps
energy diagram
vacuum
E_C
mobility edge
shallow
deep
ΔE_t
E_V
Energy
$d \approx 1\,\mu m$
poly(S-r-DIB) thin film
convergent evidence — three independent probes of the same trap
kinetic
ISPD
mechanical
MIM stack
SMU
V / A
polymer film
$I(t) \sim t^{-n}$
low n
high n
Debye
log I
log t
HV+
Probe
V_s meter
polymer film
$V_s(t)$
τ_d
derive
t
$g(E_t)$
Shallow
Deep
E_t
V_{active}
q_{tr}
Coulomb
repulsion
$F_{Maxwell}$
electrode
air gap